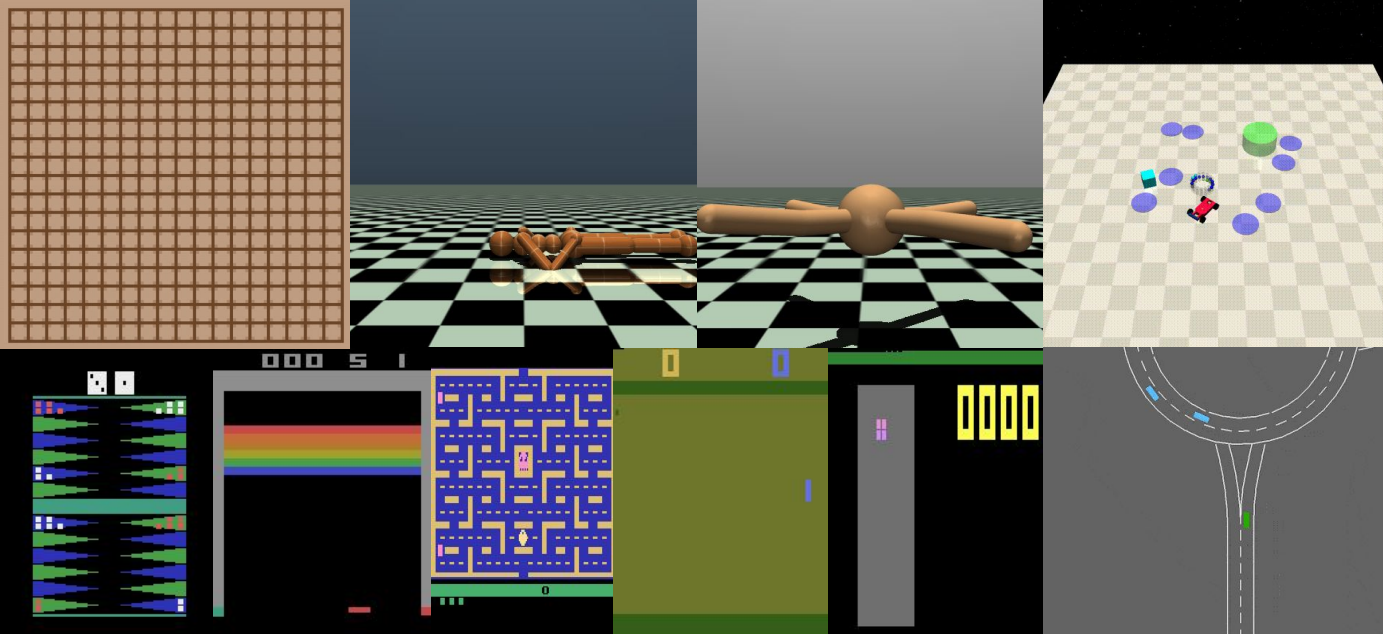




CHALLENGING PROBLEMS IN REINFORCEMENT LEARNING

Challenge Topics

Lab Course SS 2026



Artificial Intelligence
and Formal Methods

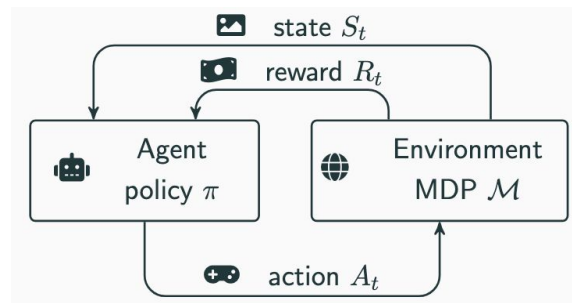
What is Reinforcement Learning?

A reinforcement learning (RL) agent

- Explores its **environment** by taking **actions** and **observing** feedback signals
- Episodically determines the optimal way to make decisions within the environment

Limitations

- Exploration is **safety-critical**
- RL is **data-hungry**
- Rewards cannot capture **sophisticated task specifications**
- RL policies are **incomprehensible and hard to implement**



Find a policy π that maximizes

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right]$$

With $\gamma \in (0, 1]$.

Challenge topics

1. **Safety**
2. **Sparse Rewards**
3. **Sim2Real**
4. **Explainability & Interpretability**
5. **Exploration vs. Exploitation**



1. Safety

What: Developing RL systems that avoid unsafe behaviors with guarantees

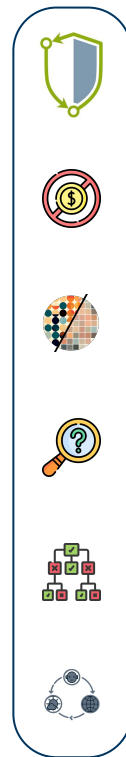
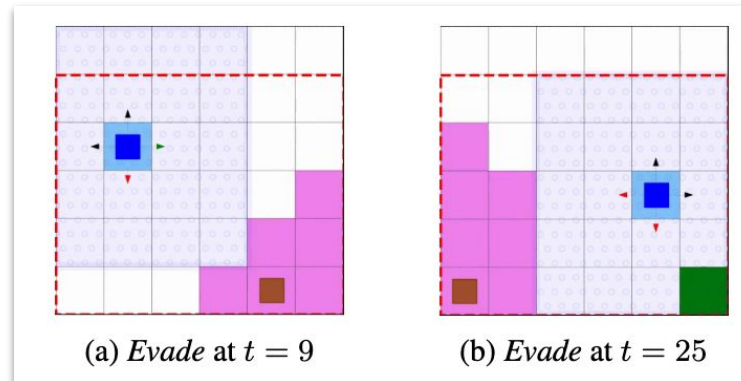
Why: Applications like autonomous driving, healthcare, robotics → risks for human life and material costs

Current approaches:

- Safe exploration → restricting the action selection during training
- Constrained RL → integrating constraints into the learning process
- Shielding → prevents agents from executing unsafe actions

Problems/questions:

- Safety vs performance: Safety does not always lead to optimal reward
- Scalability: How to apply methods to large and continuous state spaces?



2. Sparse Rewards

What: During training, the agent does not receive a reward at every time step.

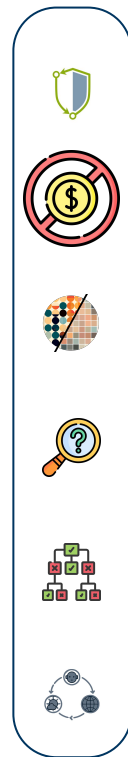
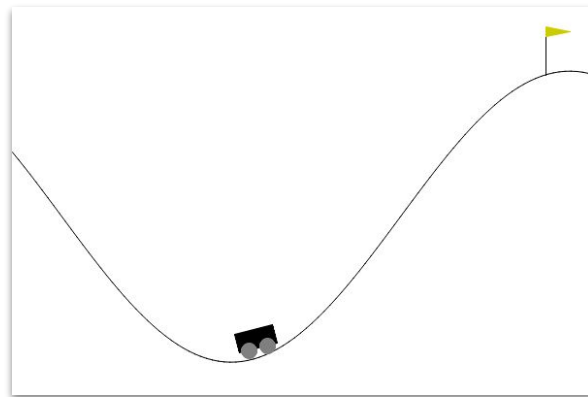
Why: Many problem settings only include reward upon task Completion, leading to large exploration space.

Current approaches:

- Reward shaping → additional feedback, hints
- Curriculum learning → easier task → harder task
- Hindsight experience replay → learning from failures
- Hierarchical RL → breaking the task down into sub-tasks

Problems/questions:

- How to determine which actions led to a good outcome? Efficient exploration?
- How to design suitable reward functions?
- How to learn robust policies?



3. Sim2Real

What: Transferring knowledge learned in simulation to real world applications

Why: Simulation vs. reality

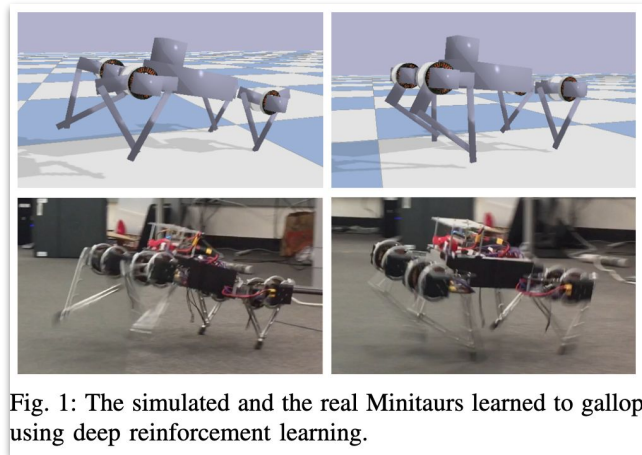
- Simulation: lower complexity, lower cost (computation + material)
- Reality: high complexity, expensive training

Common approaches:

- Domain randomization, adversarial training, meta-learning, transfer learning

Problems/Questions:

- Reality gap: controller trained on simulation might fail on real world
- Trade-off: computational complexity $\leftrightarrow$ model accuracy
- Most research is task-specific $\rightarrow$ how well does it work on other tasks?



4. Explainability & Interpretability

What: Understanding the *why* of agent's actions

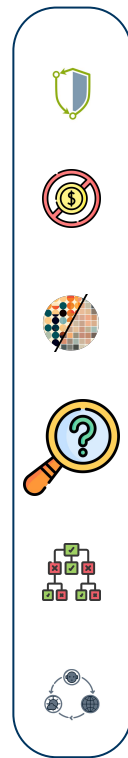
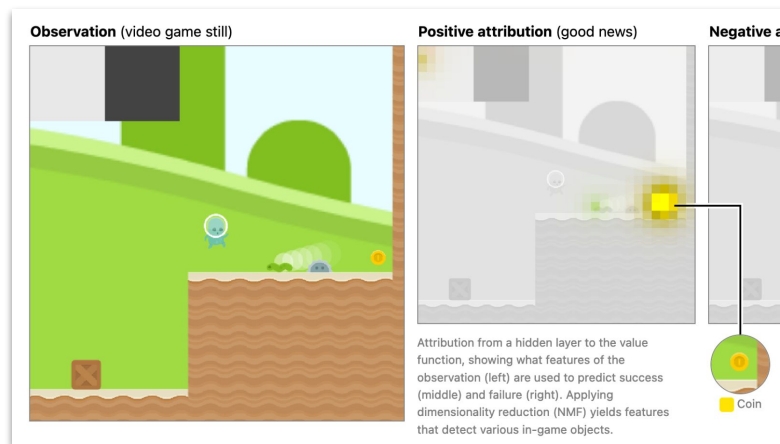
Why: Trustworthiness of AI

Current approaches:

- Attention mechanisms → which features of the environment does the agent focus on?
- Feature attribution → how do input features contribute to action selection?
- Reward decomposition → making the reward function interpretable

Problems/questions:

- RL policies are complex and nonlinear → hard to interpret
- History of sequential decisions makes it difficult to pinpoint effects of individual actions
- Performance ↔ interpretability: simple vs. complex models



5. Exploration vs Exploitation

What: Fundamental RL trade-off: exploring new actions vs relying on known policy

Why:

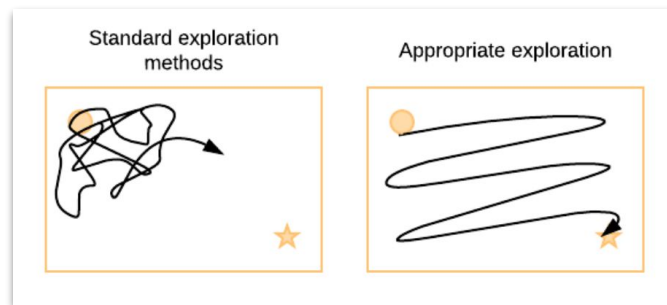
- Too much exploitation → sub-optimal policies, no adaptation/improvement
- Too much exploration → inefficient, risky, unstable

Current approaches:

- epsilon-greedy → basic
- Upper Confidence Bound (UCB)
- Thompson Sampling → considers uncertainty
- Softmax exploration

Problems/questions:

- How to incorporate prior knowledge?
- How to reduce risks in exploration?
- How to improve the efficiency of exploration?



Group Work

- Get into Bachelor or Master groups of 2-4 students.
- Within the group choose a challenge topic.
- Fill out the Moodle form “Voting on Topic Preferences”.
- Next session: We will start a deep dive into sequential decision making.